

# MedicalQA: Extending Mentorship-Oriented Question Answering to Dementia Caregiver Support

Anonymous ACL submission

## Abstract

Family caregivers of people with dementia carry a heavy informational burden, yet general-purpose LLMs give them unreliable, uncited advice. We present MEDICALQA, an extension of the mentorship-oriented QA framework MENTORQA to the healthcare domain: a question-answering resource and multi-agent generation pipeline grounded in long-form expert geriatric lectures. [TODO: dataset size, languages/scope] We compare single-agent, LLM-chunking, RAG, and multi-agent QA-generation architectures over expert lecture transcripts, and evaluate with mentorship-focused dimensions using LLM judges validated against a human annotation pilot. [TODO: headline result: multi-agent gains, judge-human agreement] Our results show [TODO: one-sentence takeaway]. We release the dataset and framework at [TODO: repo URL].

## 1 Introduction

[TODO: Write introduction following the outline in the comments above.]

## 2 Related Work

**Mentorship-oriented QA over long-form content.** Our work directly extends MENTORQA (Bhalerao et al., 2026), which introduced mentorship-focused question answering as a distinct task: answers that provide guidance, reflection, and learning value rather than isolated facts. MENTORQA released 8,990 QA pairs derived from 180 hours of long-form mentorship videos across four languages, and compared Single-Agent, Dual-Agent, RAG, and Multi-Agent generation pipelines under controlled conditions, finding that agentic coordination consistently produces higher-quality mentorship responses, with the largest gains on complex topics and lower-resource languages. Notably,

health proved to be its most difficult territory: *Physical Health* was the lowest-scoring topic (attributed to clinically dense language) and the health categories were the smallest in the dataset, while general podcast-style mentorship content offers no safety guarantees for health advice. Its authors explicitly invite extension of the open-source framework to new domains with expert oversight. MedicalQA answers this call: we retain the methodological backbone—transcript chunking, multi-agent QA generation, and mentorship-oriented evaluation—but ground it in vetted expert geriatric lectures, target dementia caregivers as a concrete, vulnerable audience, and adopt the trustworthiness requirements that the healthcare setting demands.

**Dementia caregivers’ unmet informational needs.** NLP research on dementia has concentrated on detection and diagnosis; a recent systematic review finds caregiver support to be one of the least explored directions in the field (Peled-Cohen and Reichart, 2025), even as the dementia-care community identifies LLMs as a promising avenue for caregiver assistance (Treder et al., 2024). The need itself is well documented. Analysis of over 60,000 posts on the ALZConnected forum shows caregivers seeking advice on symptoms, medications, and coping in anonymous, unverified peer spaces (Kim et al., 2024). Interview studies report caregivers improvising support from generic tools such as ChatGPT, Reddit, and online forums, while explicitly asking for *credible information with transparent sourcing* that adapts to their stage in the caregiving journey (Shi et al., 2025). A recent taxonomy triangulating 52 studies with caregiver interviews pinpoints the misalignment: existing informational resources deliver symptom facts rather than the skill-building and tailored guidance caregivers need, and current AI chatbots offer shallow, untailored support (Wang et al., 2026). Re-

mote monitoring technologies, though frequently justified by caregiver burden, likewise leave caregivers’ informational needs unaddressed (Shaik et al., 2025). These studies identify requirements but build no system that meets them.

**Curated caregiver interventions.** The accepted non-AI intervention model confirms that expert-authored, digitally delivered education helps caregivers—but does not scale. The WHO’s iSupport programme packages expert caregiving knowledge into a five-module self-help curriculum adapted across many countries (World Health Organization, 2019); yet it is static: caregivers cannot ask their own questions, and each cultural adaptation requires manual expert effort. A meta-analysis of computerized cognitive training quantifies the underlying scalability dilemma: professionally supervised digital training is effective (verbal-memory SMD 0.72) while unsupervised self-administered versions lose most of the effect (0.21), leading the authors to call explicitly for incorporating AI into self-administered interventions (Chan et al., 2024). Expert knowledge helps, but only when a professional is in the loop—motivating systems that deliver expert-level guidance interactively and at scale.

**LLMs as caregiving advisors.** Pre-LLM conversational support for dementia was scarce and shallow: a systematic review found only 6 of 505 candidate chatbots relevant, none grounding their health content in evidence (Ruggiano et al., 2021). The LLM era has not closed the reliability gap. When out-of-the-box models were evaluated against professional gold standards for home-care instruction, even GPT-4o omitted relevant details in 60–80% of scenarios and matched professional-level specificity in only 10% of responses (Pérez-Esteve et al., 2025). Caregivers probing a plain GPT-4o chatbot valued on-demand access but distrusted its uncited, generic answers; the study’s foremost design recommendation is trustworthy information with transparent source citation (Shi et al., 2026). Injecting curated domain knowledge helps—prompt engineering over 90 hand-written QA pairs significantly improved actionability, relevance, and satisfaction (Zhou et al., 2026; Oh et al., 2025)—but the authors themselves flag manual prompt curation as unscalable. Beyond content quality, the NurValues benchmark shows that even frontier LLMs trail human nurses in aligning with professional care values under adversarial dia-

logue, and that medical fine-tuning alone does not fix this (Yao et al., 2025). The verdict across these evaluations is consistent: caregiver-facing LLM answers must be grounded in vetted expert knowledge, and the grounding must scale beyond hand-curated prompts.

**Grounded and domain-specific care systems.** The systems closest to ours ground generation in curated care knowledge. CaLM combines RAG with fine-tuning over caregiving web resources, showing a 7B domain model can outperform GPT-3.5 for ADRD caregiving answers (Parmanto et al., 2024). RISE couples RAG over the evidence-based REACH Caregiver Notebook with a social robot, reaching 87% backend correctness in a small user study (Morris et al., 2026). In the professional-care domain, NurseLLM demonstrates the value of specialization—and its multi-agent variant adds a further +3.62% accuracy over the single model, independently validating agentic designs for care QA (Khondaker et al., 2025). Yet every knowledge source in this line of work is short-form written material: web pages, a structured notebook, synthetic exam questions, or hand-written QA pairs. None taps long-form expert instruction such as lecture video; none uses multi-agent coordination for question generation and answer synthesis; none frames the interaction as mentorship; and none pairs scalable LLM-as-judge evaluation with human annotation.

**Positioning.** Table 1 situates MedicalQA. Against the grounded caregiving systems, we contribute MENTORQA’s multi-agent generation and mentorship-oriented evaluation; against MENTORQA, we contribute a vetted expert knowledge source (long-form geriatric lectures), a healthcare domain and audience, and the safety bar that domain imposes.

### 3 The MedicalQA Dataset and Pipelines

#### 3.1 Expert Lecture Corpus

[TODO: Corpus description: videos, hours, transcription, preprocessing.]

#### 3.2 QA-Generation Architectures

[TODO: Describe the four pipelines and the five multi-agent roles (architect, inquisitor, scorer, justifier, synthesizer); base model and serving setup (Qwen weights on the cluster).]

|                  | MentorQA            | CaLM                 | RISE             | NurseLLM            | MedicalQA (ours)                |
|------------------|---------------------|----------------------|------------------|---------------------|---------------------------------|
| Knowledge source | Mentorship podcasts | Caregiving web pages | REACH notebook   | Synthetic MCQs      | Expert geriatric lectures       |
| Domain           | General mentorship  | ADRD caregiving      | ADRD caregiving  | Nursing exams       | Dementia caregiving             |
| Grounding        | RAG / segment       | RAG + fine-tune      | RAG              | Fine-tune only      | RAG over lecture chunks         |
| Generation       | Multi-agent         | Single model         | Single model     | MA answer selection | Multi-agent QA gen. + synthesis |
| Framing          | Mentorship          | Q&A                  | Education        | Exam prep           | Mentorship-style education      |
| Evaluation       | LLM judges + human  | Auto metrics         | Small user study | MCQ accuracy        | LLM-judge + human pilot         |

Table 1: MedicalQA compared with its foundation (MENTORQA) and the closest grounded or domain-specific care systems.

### 3.3 Grounding and Quality Control

[TODO: Segment-grounded answers, quota/scoring if applicable, PII and safety considerations for health content.]

## 4 Evaluation Methodology

[TODO: Write evaluation methodology.]

## 5 Results

[TODO: Results tables/figures and analysis.]

## 6 Conclusion

[TODO: Write conclusion.]

## Limitations

[TODO: Write limitations.]

## Ethical Considerations

[TODO: Write ethical considerations.]

## References

- Parth Bhalerao, Diola Dsouza, Ruiwen Guan, and Oana Ignat. 2026. [Beyond factual QA: Mentorship-oriented question answering over long-form multi-lingual content](#). *Preprint*, arXiv:2601.17173.
- Aaron T. C. Chan, Roy T. F. Ip, Joshua Y. S. Tran, Joyce Y. C. Chan, and Kelvin K. F. Tsoi. 2024. [Computerized cognitive training for memory functions in mild cognitive impairment or dementia: a systematic review and meta-analysis](#). *npj Digital Medicine*, 7:1.
- Md Tawkat Islam Khondaker, Julia Harrington, and Shady Shehata. 2025. [NurseLLM: The first specialized language model for nursing](#). In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 716–752, Suzhou, China. Association for Computational Linguistics.
- Jiyeong Kim, Zhuo Ran Cai, Michael L. Chen, Shawheen J. Rezaei, Sonia Onyeka, Carolyn I. Rodriguez, Tina Hernandez-Boussard, Vladimir Filkov, Rachel A. Whitmer, Eleni Linos, and Yong K. Choi. 2024. [Mental health care needs of caregivers of people with Alzheimer’s disease from online forum analysis](#). *npj Mental Health Research*, 3:54.
- Tyler Morris, Conor Brown, Xiaopeng Zhao, Linda Nichols, Jennifer Martindale-Adams, Sharon Bowland, and Wenjun Zhou. 2026. [Transforming dementia caregiver support with AI-powered social robotics](#). *Frontiers in Robotics and AI*, 12:1704313.
- Kyeong Mi Oh, Sungsoo Hong, Ziwei Zhu, Huayu Zhou, and Jung Ah Lee. 2025. [Evaluating large language models to support dementia caregivers: Identifying opportunities for improvement](#). *Innovation in Aging*, 9(Supplement 2). Conference abstract, Gerontological Society of America Annual Scientific Meeting.
- Bambang Parmanto, Bayu Aryoyudanta, Timothius Wilbert Soekinto, I Made Agus Setiawan, Yuhan Wang, Haomin Hu, Andi Saptono, and Yong Kyung Choi. 2024. [A reliable and accessible caregiving language model \(CaLM\) to support tools for caregivers: Development and evaluation study](#). *JMIR Formative Research*, 8:e54633.
- Lotem Peled-Cohen and Roi Reichart. 2025. [A systematic review of NLP for dementia: Tasks, datasets, and opportunities](#). *Transactions of the Association for Computational Linguistics*, 13:1204–1244.
- Clara Pérez-Estève, Mercedes Guilabert, Valerie Matarredona, Einav Srulovici, Susanna Tella, Reinhard Strametz, and José Joaquín Mira. 2025. [AI in home care—evaluation of large language models for future training of informal caregivers: Observational comparative case study](#). *Journal of Medical Internet Research*, 27:e70703.
- Nicole Ruggiano, Ellen L. Brown, Lisa Roberts, C. Victoria Framil Suarez, Yan Luo, Zhichao Hao, and Vagelis Hristidis. 2021. [Chatbots to support people with dementia and their caregivers: Systematic review of functions and quality](#). *Journal of Medical Internet Research*, 23(6):e25006.
- Mohammad Arif Shaik, Fahim Islam Anik, Md Mehedi Hasan, Sumit Chakravarty, Mary Dioise Ramos, Mohammad Ashiqur Rahman, Sheikh Iqbal Ahamed, and Nazmus Sakib. 2025. [Advancing remote monitoring for patients with Alzheimer disease and related dementias: Systematic review](#). *JMIR Aging*, 8:e69175.

- Jiayue Melissa Shi, Keran Wang, Dong Whi Yoo, Ravi Karkar, and Koustuv Saha. 2025. [Balancing caregiving and self-care: Exploring mental health needs of Alzheimer's and dementia caregivers](#). *Proceedings of the ACM on Human-Computer Interaction*, 9(7).
- Jiayue Melissa Shi, Dong Whi Yoo, Keran Wang, Violeta J. Rodriguez, Ravi Karkar, and Koustuv Saha. 2026. [Mapping caregiver needs to AI chatbot design: Strengths and gaps in mental health support for Alzheimer's and dementia caregivers](#). *ACM Transactions on Computing for Healthcare*.
- Matthias S. Treder, Sojin Lee, and Kamen A. Tsvetanov. 2024. [Introduction to large language models \(LLMs\) for dementia care and research](#). *Frontiers in Dementia*, 3:1385303.
- Keran Wang, Drishti Goel, Jiayue Melissa Shi, Violeta J. Rodriguez, Daniel S. Brown, Dong Whi Yoo, Ravi Karkar, and Koustuv Saha. 2026. [A taxonomy of mental health and technology needs for Alzheimer's and dementia caregivers](#). *Preprint*, arXiv:2606.19247.
- World Health Organization. 2019. iSupport: e-health training programme for caregivers of people living with dementia. <https://www.who.int/teams/mental-health-and-substance-use/treatment-care/isupport>. Accessed: 2026-08-15.
- Ben Yao, Qiuchi Li, Yazhou Zhang, Siyu Yang, Bohan Zhang, Prayag Tiwari, and Jing Qin. 2025. [Nur-Values: Real-world nursing values evaluation for large language models in clinical context](#). *Preprint*, arXiv:2505.08734.
- Huayu Zhou, Ziwei Zhu, Kyeong Mi Oh, and Sungsoo Ray Hong. 2026. [Empowering informal caregivers of persons with early-stage dementia by large language models: Mixed methods evaluation](#). *JMIR Formative Research*, 10:e79975.